

Solutions to 2020-2021-1 EI243 Final Exam

MA Dianguang, 2020/11/24

PART I (40%)

Question	1-5	6-10	11-15	16-20
Paper A	BDCAD	DCCBA	BCDBA	BCDAC
Paper B	CABDA	DBACB	BAAAB	CBADB

PART II (36%)

Question	Paper A	Paper B
21	AND	AND
22	86	101
23	1011 0001.1101	1010 0111.1101
24	-79	-93
25	9C28	9C2A
26	$\overline{ABD}$	$\overline{ABD}$
27	$\overline{\overline{A+B+C} + \overline{A+B+C} + \overline{A+B+C}}$	$\overline{\overline{A+B+C} + \overline{A+B+C} + \overline{A+B+C}}$
28	$\overline{BC} + \overline{BD}$	$\overline{BC} + \overline{BD}$
29	1	0
30	212	212
31	3	3
32	10, 01	10, 01

PART III (24%)

33 (9%)

Denote the inputs as A , B , and C , and the output as X . From the problem description, we can develop the K-map for X as shown in Figure 33(a), and then the minimized expression

$$X = AB + BC + CA$$

The logic diagram is shown in Figure 33(b).

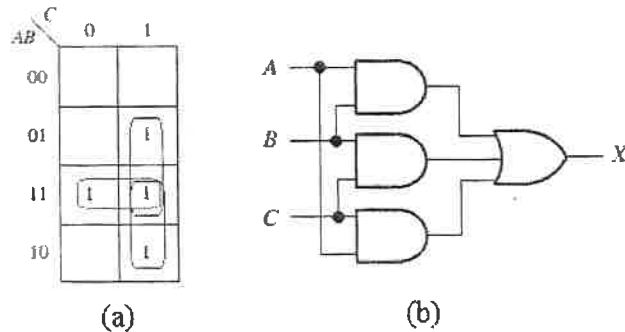


Figure 33

34(15%)

From the problem description, we can develop the combined K-map for the next states, as shown in Figure 34(a).

The minimized state equations are

$$Q_2^{n+1} = Q_1 Q_0, Q_1^{n+1} = Q_1 \bar{Q}_0 + \bar{Q}_1 Q_0 = Q_1 \oplus Q_0, Q_0^{n+1} = \bar{Q}_2 \bar{Q}_0$$

Hence, the excitation equations are

$$D_2 = Q_1 Q_0, D_1 = Q_1 \oplus Q_0, D_0 = \bar{Q}_2 \bar{Q}_0$$

The logic diagram is shown in Figure 34(b).

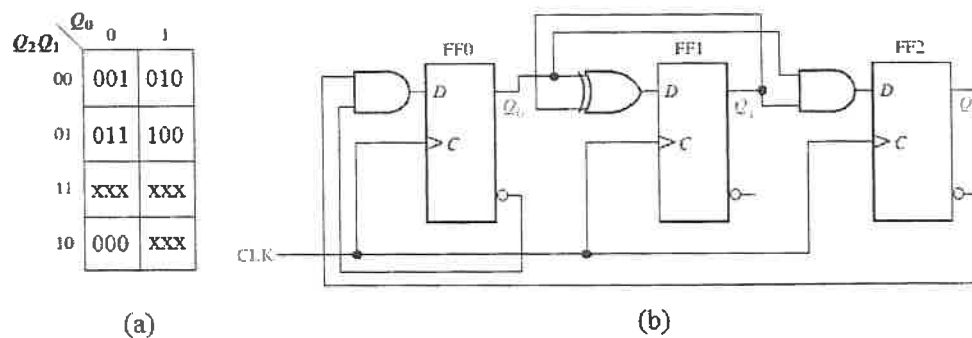


Figure 34